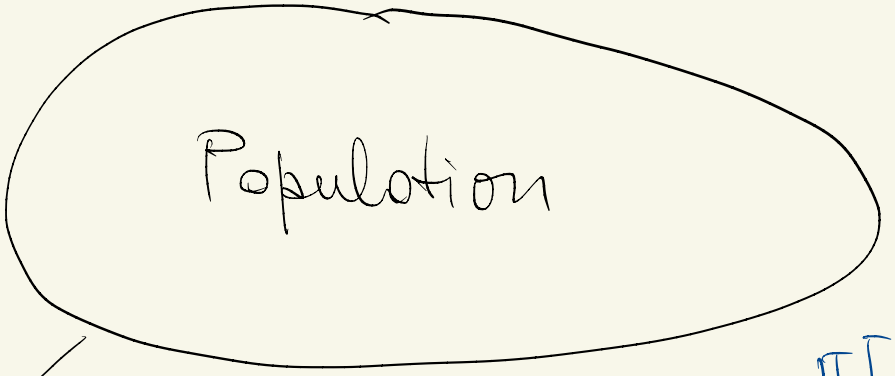
Eco 202 – Fall 2026

Class 12



Sample → random sampling + ($X_1, X_2, \ldots, X_n$, independent)

$\mathbb{E}[X_i] = \mu$

$\mathbb{V}[X_i] = \sigma^2$

the same for all $i = 1, 2, \ldots, n$

$(X_1, X_2, \ldots, X_n)$ ⟹ statistics

$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$

realization ⇓ (sample size)

⇓ realized

$(x_1, x_2, \ldots, x_n)$ ⟹ realized statistic

$\bar{x}_n = \frac{1}{n} \sum_{i=1}^{n} x_i$

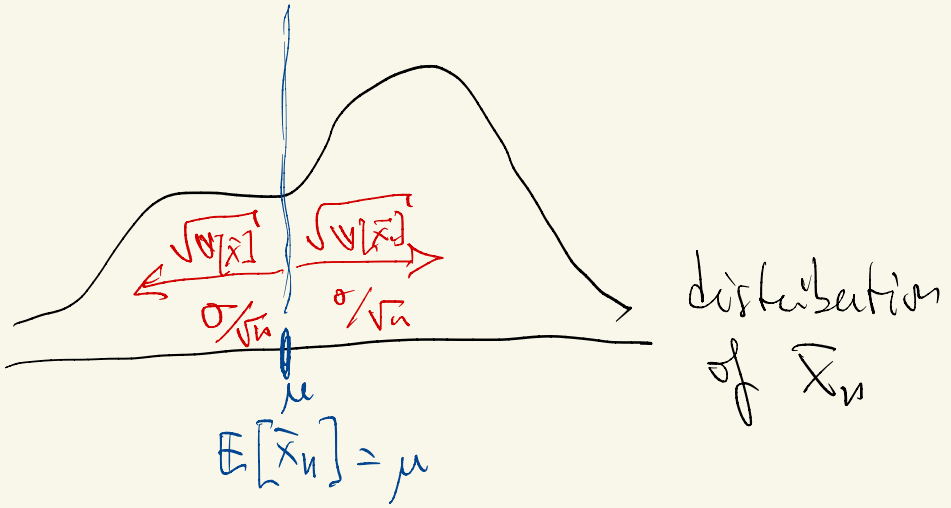
these are the numbers you see in your data

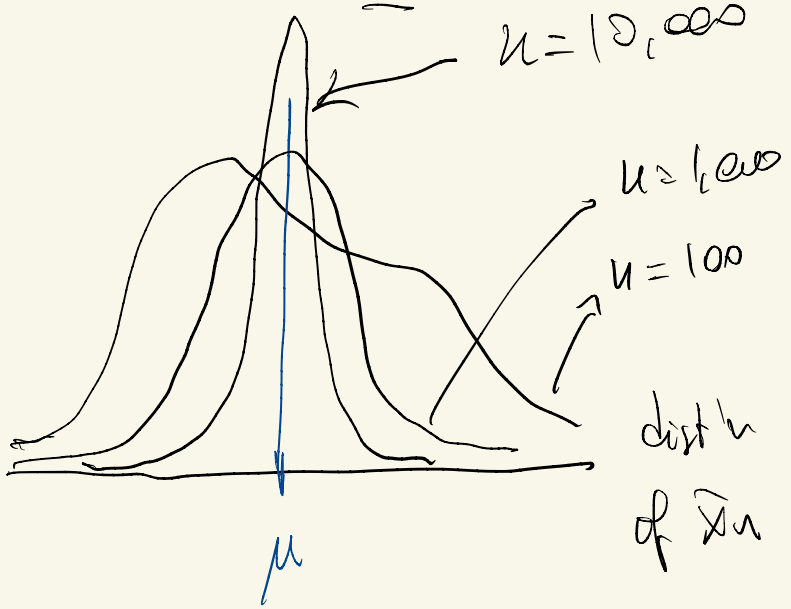
$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ is random and thus has a sampling distribution

- Properties

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n} \mu = \mu$$

$$\mathbb{V}[\bar{X}_n] = \frac{1}{n^2}\sum_{i=1}^{n} \mathbb{V}[X_i] = \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 = \frac{\sigma^2}{n}$$

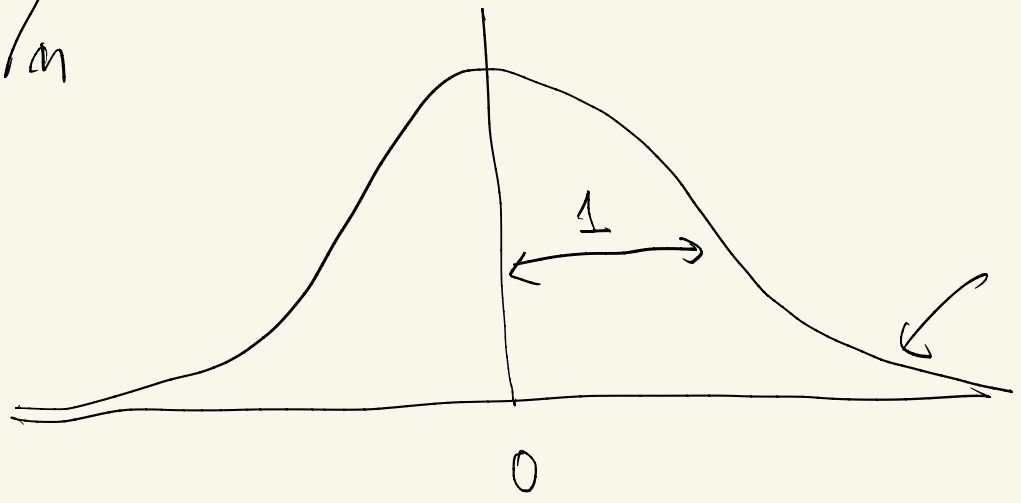




$$Z_n = \frac{\bar{X}_n - \mathbb{E}[\bar{X}_n]}{\sqrt{\mathbb{V}[\bar{X}_n]}}$$

its sampling distribution will look more and more like

$$= \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$



standard normal distribution

as $n \to \infty$!

this is called the central limit theorem

Summary: $X_1, X_2, \dots, X_n$ are iid

$\mathbb{E}[X_i] = \mu$, $\mathbb{V}[X_i] = \sigma^2$ independent and identically dist'd

- **Law of Large Numbers (LLN)**

$$\bar{X}_n \xrightarrow{P} \mu \iff \lim_{n\to\infty} \mathbb{P}\left[|\bar{X}_n - \mu| > \varepsilon\right] = 0$$

(for all $\varepsilon > 0$)

- **Central Limit Theorem (CLT)**

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1) \iff$$

sampling distribution of Z_n $\longrightarrow$ Standard Normal Distribution

Recall

$$Y \sim N(\mu, \sigma^2) \iff \frac{Y-\mu}{\sigma} \sim N(0,1)$$

thus, by CLT,

$$\bar{X}_n \overset{\cdot}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \iff \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} \overset{\cdot}{\sim} N(0,1)$$